Embeddings

Text → "what is the cost of solar panels"

Input (what is the cost of solar panels"
16 20 11 5 4 10 6
T1 T2 T3 T4 T5 T6 T7



What → 16 → numerical representation
↳ semantic meaning

Embedding

cost → 5 ⇒ [-0.1, 0.2, 0.3, 0.6] ← semantic meaning ← meaning

Embedding

Embeddings → meaningful vector representation of a word or token

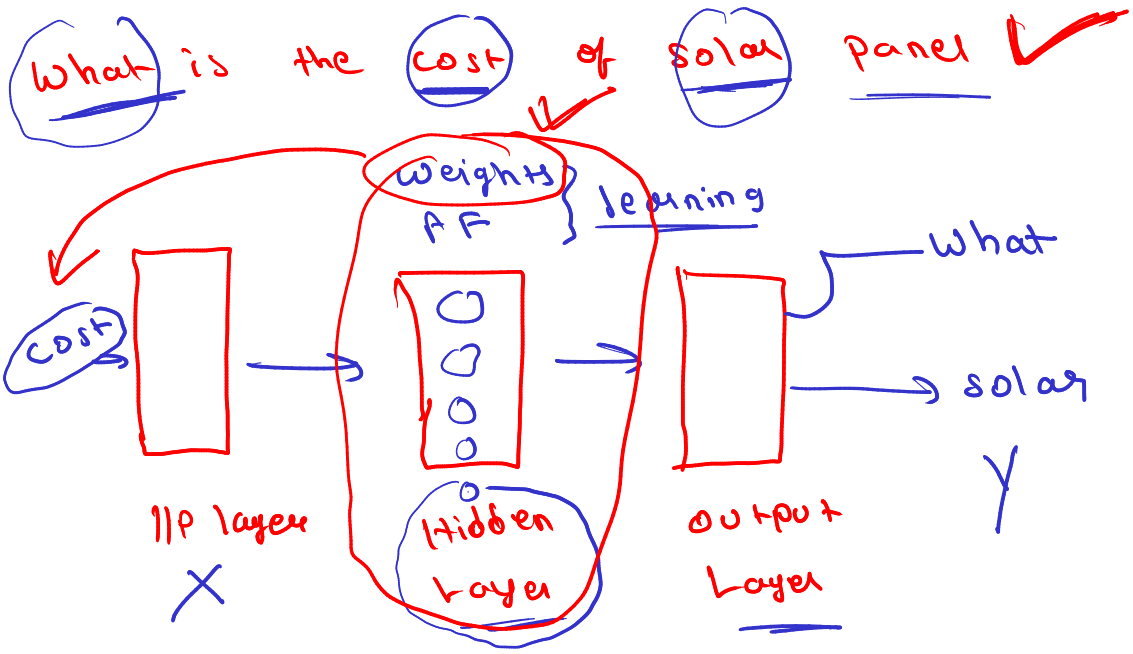
⇒ NN - neural Networks

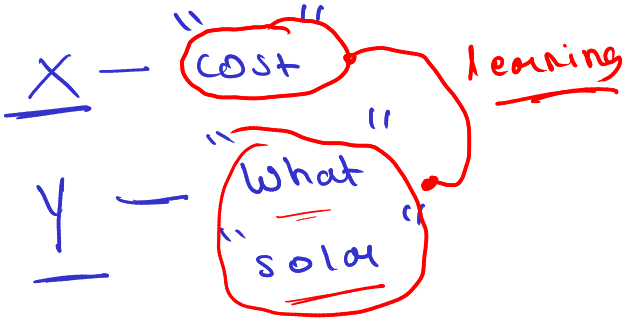
Embedding

⇒ 1. word2vec } word to vector
2. Glove

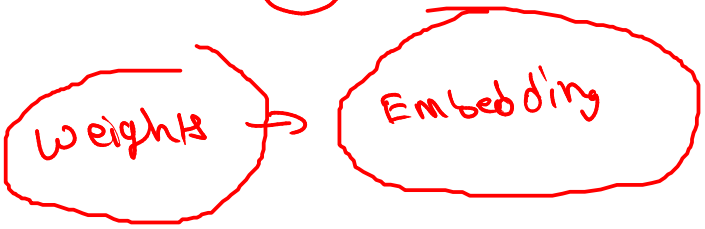
NN

⇒ skip gram
→ continous BOW

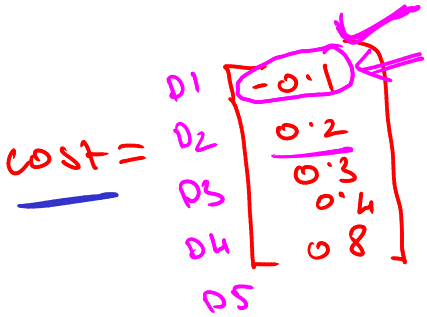




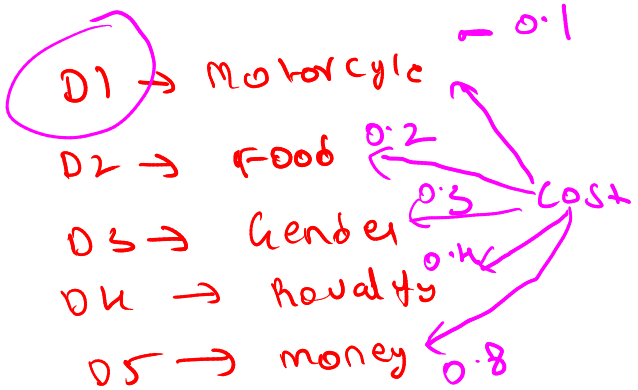
$$Y = W \cdot X + b$$

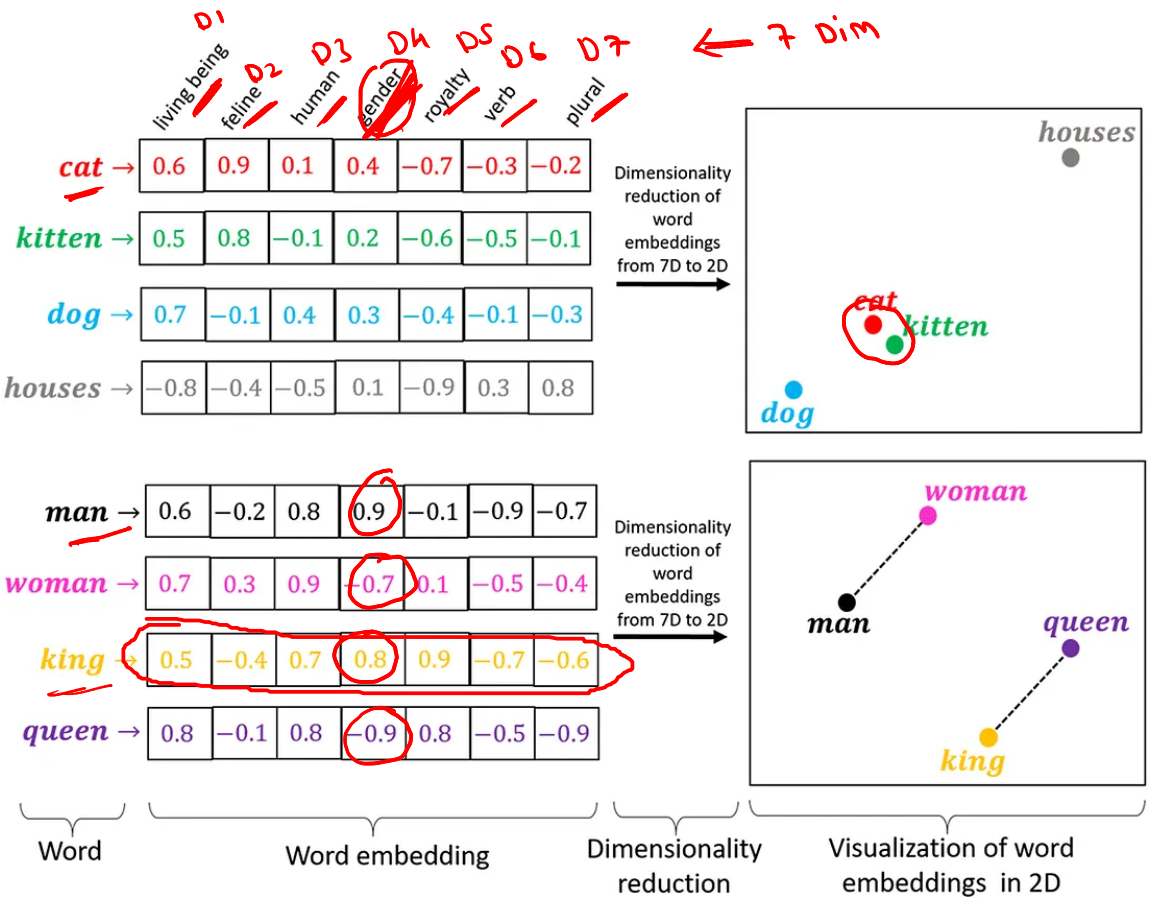


Embedding → Dimension ⇒ no of weights



Dimension





King

Queen

man

women

Word $\begin{bmatrix} D1 \\ D2 \\ \vdots \\ D7 \end{bmatrix}$

King − man + women ≈ queen

King ≈ queen

man ≈ women

Sentence2

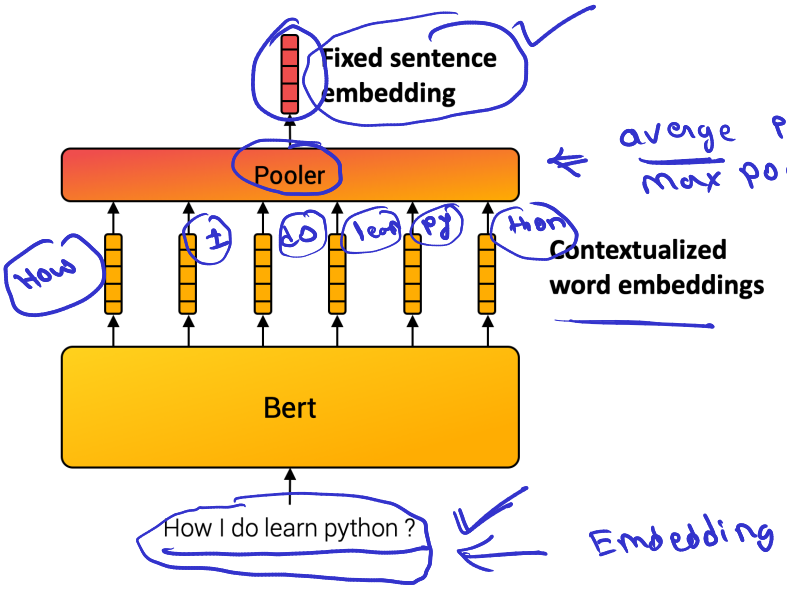
Sentence Embedding

1. BERT
2. SBERT
3. Transformer

BERT → Bidirection Encoder representation from Transform

Sentence transformer

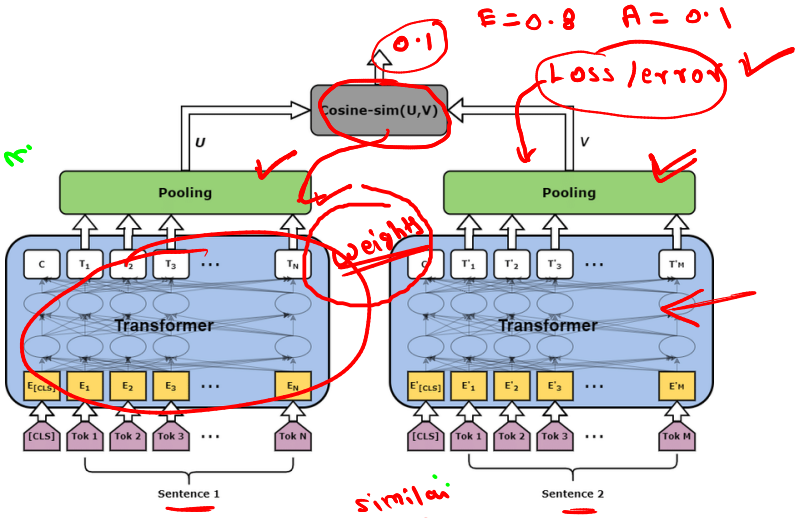
"How do I learn python" → [ ]



average pooling
max pooling

Embedding

Siamese Network



E=0.8 A=0.1

0.1

Loss/error

weight

similar

⇒ How do I learn python? (S1) ⟷ Python is easy language (S2)

⇒ AI is the future (S1) ⟷ I love coffe (S2) ⇒ 0.2

non similar